Jin Hong Moon

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Education

Johns Hopkins University — Expected Dec 2026
B.S./M.S. in Computer Science, Cumulative GPA: 3.88 — *Baltimore, MD*

Relevant Courses: Machine Learning, Deep Learning, Artificial Intelligence, Object-oriented Engineering Algorithms, Computer Systems, Databases, Linear Algebra, Probability, Statistics, Graphics
Teaching Assistant: Data Structures (Java), Intermediate Programming (C/C++)

Experience

Software Engineering Intern — May 2026 – August 2026
Google — *Sunnyvale, CA*

- Optimized Gmail SHA-256 body hashing using a C++ caching layer within DKIM/ARC signing and verification
- Reduced body hashing CPU by 69.3% for outbound and 50.8% for inbound traffic, saving 256µs per redundant pass and projecting 531 CPU cores saved at a peak scale of nearly 1M QPS

Machine Learning Researcher — Jan 2026 – Present
Johns Hopkins University — *Baltimore, MD*

- Designed an energy-based vision transformer that jointly refines feature and attention map via energy minimization, improving 1.4 mIoU over a baseline with near-zero added parameters
- Built a cross-layer feedback system that propagates converged attention states between transformer layers

Software Engineering Intern — June 2025 – August 2025
Bloomberg — *New York City, NY*

- Designed a distributed, scalable data pipeline for PORT⟨GO⟩ time-series data layer, developing a 20+ threaded C++ producer, Python consumer, and a 4-partition Confluent Kafka topic
- Optimized read query performance of a C++ high-throughput, low-latency infrastructure processing 4B+ daily queries at 25K requests/sec, reducing average latency by 32.6%, significantly accelerating PORT⟨GO⟩ analytics
- Built a real-time time-series analytics layer using ANSI SQL, Apache Trino, and Superset

Software Engineering Intern — June 2024 – November 2024
Claudius Legal Intelligence — *Remote*

- Developed 5+ API endpoints integrated with jQuery and AJAX frontends for seamless pagination and filtering
- Initiated AI legal assistance by implementing locally fine-tuned BERT and BART models with PEFT LoRA in PyTorch, generating trending article topics and recommendations

Projects

Summit - Job Application Tracker | summit-jobs.com — February 2025 – May 2025

- Created a distributed task queue using Celery and Redis in Django REST Framework to parse job-related content from Gmail via OpenAI API with 92.6% accuracy, enabling automated job tracking from users' inboxes
- Improved a lightweight DistilBERT-based classifier to filter non-job-related emails before parsing with accuracy of 96.5%, significantly reducing token usage and improving throughput to 5K tokens/sec
- Built an OpenAI-powered chatbot with semantic search over 16K job postings by leveraging vector embeddings generated via text-embedding-3-small and stored in Supabase's pgvector, delivering personalized recommendations

Noori AI - Medical Translation | nooriai.com — February 2025 – May 2025

- Led development of a HIPAA-compliant sub-500ms medical interpreter application across 3 languages, reducing interpretation costs by over 90% for healthcare institutions serving patients with Limited English Proficiency
- Achieved 37.0 BLEU score on a MarianNMT model trained on OPUS and web-scraped medical data, preprocessed with custom tokenizers to produce aligned sentence pairs, under the mentorship of Dr. Philipp Koehn

Technical Skills

Languages: Python, C, C++, Go, Java, SQL
Frameworks: Flask, Django, FastAPI, Spring Boot, Supabase, Node.js, React.js, Next.js
Developer Tools: Docker, Kubernetes, CMake, AWS, GCP, Apache Kafka, Apache Spark, Apache Trino
Libraries: PyTorch, TensorFlow, Scikit-learn, HuggingFace, Numpy, Pandas, Celery, gMock